$\mathcal{P}_{\text{clu}}$ Context-aware Logical Units Clustering

Instruction:
You are given a sequence of reasoning units...
[*Logical Units Template*]

Your task is to cluster consecutive reasoning units that are semantically connected...

Expected Output Format: [*Output Guideline*]

Output: {"s_0": {"title":..., "content":...},...}

Figure 3: Our abbreviated prompt template to guide LLM to cluster reasoning units into logical cohesive reasoning steps. For detailed *Logical Units Template* & *Output Guideline* see Appendix A.

code generation, M can be excessively large, resulting in an unwieldy number of fragmented units that hinder semantic analysis and dependency extraction. Moreover, the model's stylistic tendency to insert delimiters frequently can lead to reasoning units that are too fine-grained, often lacking coherent context for standalone analysis.

Context-aware Logical Units Clustering. To address these challenges, we introduce a context-aware logical units clustering procedure that aggregates semantically related reasoning units into higher-level *reasoning steps*. Specifically, we leverage a large language model (LLM) to sample possible clusterings under decoding temperatures τ_r conditioned on a carefully designed prompt template $\mathcal{P}_{\text{clu}}$ (see Figure 3) $\mathcal{P}_{\text{clu}}$:

$$S = (s_1, s_2, \ldots, s_K) \sim P_{\text{LLM}}(S \mid \mathcal{P}_{\text{clu}}, U; \tau_r),$$

where each s_j is ideally formed by concatenating adjacent u_i meeting a semantic affinity criterion, with $K \ll M$. This aggregation aims to ensure that each reasoning step s_j provides sufficient context for downstream analysis while maintaining a manageable total number of segments.

Yet, given the generative nature of LLMs, repeated invocations of the clustering prompt do not guarantee an identical clustering. Rather than treating this variability as noise, we further harness it through a further ensemble sampling and selection approach to identify the most coherent clustering.

Ensemble Sampling To capture the full range of possible clustering, we generate an ensemble of B candidate instances of clustering by independently sampling $P_{\text{LLM}}(S \mid \mathcal{P}_{\text{clu}}, U)$ from with varied random seeds and decoding temperatures:

$$\mathcal{C} = \{C^{(1)}, C^{(2)}, \ldots, C^{(B)}\}$$
$$C^{(b)} = (s_1^{(b)}, \ldots, s_{K_b}^{(b)})$$

Each $C^{(b)}$ is a candidate clustering of the original reasoning units.

To objectively compare these candidates, we introduce a weighted average quality score that evaluates key properties of each clustering. In general, we compute

$$F(C^{(b)}) = \sum_{\ell=1}^{L} w_\ell \, \phi_\ell(C^{(b)}),$$

where each ϕ_ℓ is a quantitative metric and the weights w_ℓ are non-negative and sum to one. In our experiments, we consider three intuitive criteria:

Criteria 1 **Intra-step coherence.**

$$\phi_{\text{ic}}(C^{(b)}) = \frac{1}{K_b} \sum_j \frac{2 \sum_{u<v \in s_j^{(b)}} \cos(\mathbf{e}_u, \mathbf{e}_v)}{|s_j^{(b)}|(|s_j^{(b)}| - 1)}$$

which rewards semantic similarity among units within each reasoning step.

Criteria 2 **Step-to-step separation.**

$$\phi_{\text{sep}}(C^{(b)}) = \frac{1}{K_b - 1} \sum_{j=1}^{K_b - 1} \left[1 - \cos(\mathbf{e}_{s_j^{(b)}}, \mathbf{e}_{s_{j+1}^{(b)}}) \right]$$

which encourages semantic distinctiveness between adjacent reasoning steps.

Criteria 3 **Length regularity.**

$$\phi_{\text{len}}(C^{(b)}) = 1 - \left| \frac{\frac{1}{K_b} \sum_j |s_j^{(b)}|}{\mu_{\text{ref}}} - 1 \right|$$

where μ_{ref} is a referenced average step length derived from the original CoT structure (see Appendix E for details), penalizing pathological segmentations that are too short or too long.

Here, $\mathbf{e}_\cdot$ denotes a sentence embedding vector derived from a pretrained encoder.

The final segmentation is selected as the candidate with the highest composite score:

$$V := C^* = \arg\max_{C \in \mathcal{C}} F(C).$$

$\mathcal{P}_{\text{sem}}$ Adjacency Matrix Sampling

Instruction:
Given an ordered sequence of K reasoning steps,...
[*Logical Steps Template*]

...Your task is to decide whether step i supports, contradicts, or is independent of step j...

Expected Output Format: [*Output Guideline*]

Output: $\{(s_0, s_1)$:..., (s_0, s_2): ..., (s_1, s_0): ...$\}$

Figure 4: Our abbreviated prompt template to detect semantical relationship between two different reasoning steps. For detailed input/output template and intuition behind the instruction, see Appendix A

This ensemble–scoring framework offers a flexible yet principled way to select coherent and analyzable reasoning step from all plausible instances of clustering generated by the LLM. The complete clustering and selection algorithm is summarized in Appendix D.

For all subsequent analyses, each s_j in the selected C^* is treated as a node in our reasoning graph, providing a compact yet semantically rich foundation for structural and dependency analysis.

3.3 Extracting Inter-Dependencies between Reasoning Steps

our next objective is to construct a directed semantic graph $G = (V, E)$, where each edge $(i, j) \in E$ represents an inferred relationship—such as support or contradiction—between step s_i and step s_j. To achieve this, we propose a rejection sampling-based semantic detection procedure that fuses global predictions from multiple LLM samplings.

Adjacency Matrix Sampling. We first obtain diverse global views of step-wise dependencies by repeatedly prompting the LLM with a structured template $\mathcal{P}_{\text{sem}}$ (see Figure 3). Each prompt presents the entire ordered set of reasoning steps and requests predictions for every ordered pair (i, j), $i < j$. The model outputs a full adjacency matrix:

$$A^{(r)} \sim P_{\text{LLM}}\left(A \mid \mathcal{P}_{\text{sem}}, V; \tau_r\right)$$
$$A^{(r)} \in \{-1, 0, 1\}^{K \times K},$$

where $A^{(r)}_{ij} = 1$ (support), -1 (contradict), or 0 (independent). We repeat this sampling R times, varying the decoding temperature τ_r to enhance diversity. This strategy ensures the LLM can leverage the full context for globally consistent predictions while capturing uncertainty.

Adaptive Edge-wise Probability Estimation. For each possible edge (i, j), we aggregate predictions across the R samples to estimate the empirical probability of each relation:

$$\hat{p}_{ij}(c) = \frac{1}{R} \sum_{r=1}^{R} \mathbf{1}\left[A^{(r)}_{ij} = c\right], \qquad c \in \{-1, 0, 1\}.$$

These aggregated probabilities provide a measure of confidence for the existence and type of each possible semantic relation.

Rather than relying on a fixed number R of sampled adjacencies, we fuse information across all samples for a robust final graph construction. Sampling continues until the estimated probabilities for all edges reach a specified confidence level. Specifically, for each edge, we compute the pooled standard error:

$$\mathrm{SE}_{ij} = \sqrt{\frac{1}{R} \sum_{l \in \{-1,+1\}} \hat{p}_{ij}(l)\big(1 - \hat{p}_{ij}(l)\big)}$$

Here, we explicitly omitted ($l = 0$) case to simplify the estimator while preserving the accuracy required for the adaptive stopping criterion (see Appendix F). The process halts once $\max_{i<j} \mathrm{SE}_{ij} \leq \varepsilon$ [1], or a hard cap $R_{\max}$ is reached. This guarantees that our edge probability estimations are statistically reliable before moving to the next phase.

With reliable probability estimates, we next construct the final adjacency matrix via a consensus rule. For each pair (i, j), we define the signed confidence:

$$w_{ij} := \hat{p}_{ij}(+1) - \hat{p}_{ij}(-1), \quad w_{ij} \in [-1, 1].$$

We then apply a dual-threshold criterion:

$$A_{ij} = \begin{cases} +1, & \text{if } w_{ij} \geq \tau_{\text{pos}} \\ -1, & \text{if } w_{ij} \leq -\tau_{\text{neg}} \\ 0, & \text{otherwise,} \end{cases} \qquad A_{ji} = -A_{ij},$$

where τ_{pos} and τ_{neg} can be tuned independently[2].

The resulting weighted adjacency $W = \left[w_{ij}\right]_{i,j}$ is preserved for future analysis, while we pay attention to the hard-thresholded $A = \mathrm{sign}(W) \odot \mathbf{1}[|W| \geq \tau]$, which serves as the binary backbone for structural analysis. We also include a complete

[1] in practice $\varepsilon = 0.05$ suffices

[2] We typically set $\tau_{\text{pos}} = 0.4$, $\tau_{\text{neg}} = 0.3$ to reflect the empirical imbalance between supporting and contradicting links.

adjacency matrix sampling and adaptive edge estimation algorithm in Appendix D.

To sum up, the pipeline described in this section provides a principled and systematic framework for converting raw reasoning traces from RLMs into interpretable graph structures. This unified graph representation serves as a powerful analytical tool, enabling fine-grained examination of how RLMs organize, connect, and validate intermediate inferences. In the following sections, we will leverage this reasoning graph formalism to quantitatively analyze the internal reasoning dynamics and decision-making behaviors of advanced RLMs.

4 Reveal Cognitive Behavior of RLM with Reasoning Graph

Existing analysis of Reasoning LLMs (RLMs) have primarily relied on performance-based metrics such as accuracy or token-level statistics like reasoning length. While these measures offer a coarse understanding of model behavior, they fail to capture the complex and dynamic structure exposed by RLM's output.

In this section, we propose to move beyond token-level perspectives and instead leverage the reasoning graph constructed in Section 3 as an effective medium for cognitive analysis. By representing the model's chain-of-thought as a graph of semantically coherent reasoning steps (nodes) and their directed relationships (edges), we can systematically quantify the structure, flexibility, and effectiveness of model reasoning. This shift enables us to answer deeper questions about how RLMs organize, explore, and consolidate information during problem problem-solving process. Figure 2 provides a concrete example of RLM's reasoning process for solving a spatial geometric problem.

All implementation details are included in Appendix C.

4.1 Graph-Based Metrics for Quantifying Model Reasoning

To systematically analyze the cognitive organization of RLM reasoning, we introduce several graph-based metrics, each designed to capture a distinct structural aspect of the reasoning process.

Exploration Density (ρ_E): Measures the overall connectivity among reasoning steps, reflecting the breadth of the model's exploration.

$$\rho_E(G) = \frac{|E|}{|V|(|V|-1)}$$

Higher values indicate denser intra-reasoning-step correlations.

Branching Ratio (γ_B): Quantifies the diversity of alternative reasoning paths, capturing the model's capacity for exploring parallel ideas and diverse solutions.

$$\gamma_B(G) = \frac{|\{s \in V \mid \mathrm{d}^-(s) > 1\}|}{|V|}$$

where $d^-(s)$ is the out-degree of node s.

Convergence Ratio (γ_C) : Captures the extent to which the model integrates multiple reasoning threads into unified conclusions, indicating its ability to synthesize disparate ideas.

$$\gamma_C(G) = \frac{|\{s \in V \mid \mathrm{d}^+(s) > 1\}|}{|V|} \tag{1}$$

where $d^+(s)$ is the in-degree of node s.

Linearity (ℓ): Represents the prevalence of strictly sequential reasoning, measuring the fraction of nodes with degree greater than one.

$$\ell(G) = 1 - \frac{|\{s \in V \mid d(s) > 2\}|}{|V|} \tag{2}$$

where $d(s)$ is the total degree of node s.

Collectively, these metrics offer a comprehensive high-level view of the reasoning graph structure. They allow us to quantify not only how much the model reasons, but how it organizes its thinking through branching exploration, convergence, or rigid linearity. In the following analyses, we will show how these quantities are directly related to and influence model performance.

4.2 Impact of Prompting Paradigms on Reasoning Structure

Having established our graph-based metrics, we next investigate how different prompting styles shape the internal reasoning structure of RLMs. We focus on three few-shot demonstration styles: `Minimal`, `Concise`, and `Explanatory`. (see Appendix B for detailed definitions). Each style provides a distinct richness of context, ranging from bare problem–answer pairs to human-like concise reasoning and extended, self-reflective chains generated by the model itself.